MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, MULTICONDUCTOR, (2 CONDUCTORS) AND SINGLE SHIELDED M27072/81E8CK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-27072.

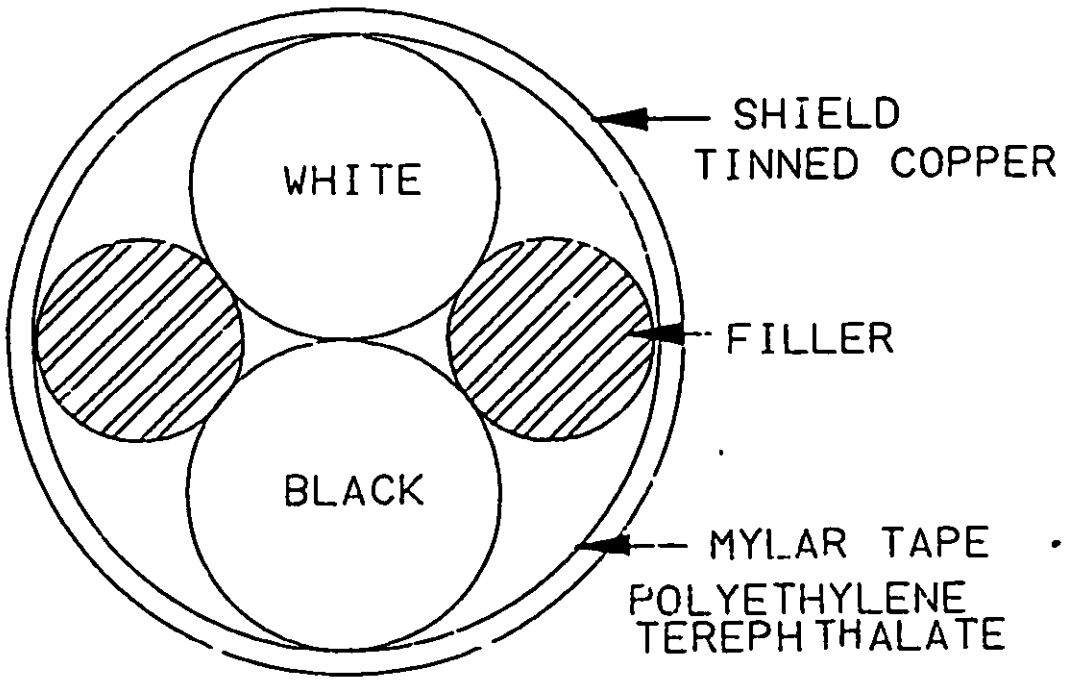


FIGURE 1. Cable configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:55Z
Check the source to verify that this is the current version before use.

DESIGN DATA:

Design and construction: See figure 1.

Number of conductors: Two.

Conductor size and stranding: See table I.

Wire: In accordance with M16878/18BKE*.

Jacket undershield: Not applicable.

Shielding: Tinned copper.

Sheath: Not applicable.

Temperature rating: 200°C.

REQUIREMENTS:

Visual and dimensional:

Components:

Basic wire: Tinned copper.

Dimension: See table I.

Primary insulation:

Material: Polyvinyl chloride.

Wall thickness: .017 inch ±.003.

Colors:

White.

Black.

Spark test voltage: 5,000 V rms.

Inspection test voltage: 3,000 V rms.

Impulse dielectric test: 10,000 V peak.

Jacket over insulation: Polyamide.

Wall thickness: .004 inch ±.001.

Color: Natural.

Filler: Applicable.

Material: In accordance with MIL-C-27072.

Lay: 3.0 inches maximum.

Tape material: Polyethylene terephthalate.

Shield:

Material: Tinned copper.

Gauge: 36 AWG.

Braid angle: 20 to 40 degrees.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:55Z
Check the source to verify that this is the current version before use.

Workmanship: In accordance with MIL-C-27072.

Finished cable:

Diameter: In accordance with table I.

Conductor resistance: In accordance with MIL-W-16878/18.

Dielectric strength: In accordance with MIL-W-16878/18.

Cold bend: Mandrel size in accordance with table XIII of MIL-C-27072.

Ozone: Mandrel size in accordance with table XIV of MIL-C-27072.

Marking durability: In accordance with MIL-C-27072.

Heat stability: Not applicable.

TABLE I. Construction data.

Part number M27072/81	Conductor				Finished cable diameter (in.) ±10%	Basic wire M16878/18
	Wire size (AWG)	Number of strands	Strand size (AWG)	Nominal diameter (in.)		
EBCK	14	19	27	.113	.252	BKE***

Asterisks (*) shall be replaced by the appropriate color designator.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Air Force - 99
DLA - ES, IS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6145-0909-39)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:55Z
Check the source to verify that this is the current version before use.